

Code:

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#include <iostream>
#include <vector>
#include <algorithm>
#include <cmath>
using namespace std;
int main() {
    int n, head, diskSize;
    char direction;
    cout << "Enter number of disk requests: "; cin >> n;
    vector<int> requests(n);
    cout << "Enter disk requests:\n";
    for (int i = 0; i < n; i++) cin >> requests[i];
    cout << "Enter initial head position: "; cin >> head;
    cout << "Enter disk size (maximum cylinder): ";
    cin >> diskSize;
    cout << "Enter direction (L for Left, R for Right): ";
    cin >> direction;
    vector<int> left, right, sequence;
    if (direction == 'L' || direction == 'l')
        left.push_back(0);
    else
        right.push_back(diskSize - 1);
    for (int req : requests) {
        if (req < head)
            left.push_back(req);
        else
            right.push_back(req); }
    sort(left.begin(), left.end());
    sort(right.begin(), right.end());
    int seekCount = 0;
    int current = head;
    if (direction == 'R' || direction == 'r') {
        for (int req : right) {
            sequence.push_back(req);
            seekCount += abs(current - req); current = req;
        }
        for (int i = left.size() - 1; i >= 0; i--) {
            sequence.push_back(left[i]);
            seekCount += abs(current - left[i]);
            current = left[i];
        }
    }
    else {
        for (int i = left.size() - 1; i >= 0; i--) {
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        sequence.push_back(left[i]);
        seekCount += abs(current - left[i]);
        current = left[i];
    }
    for (int req : right) {
        sequence.push_back(req);
        seekCount += abs(current - req);
        current = req;
    }
}
cout << "\nSeek Sequence:\n";
cout << head;
for (int x : sequence) cout << " -> " << x;
cout << "\n\nTotal Seek Distance = " << seekCount << endl;
cout << "Average Seek Distance = "
    << (float)seekCount / n << endl;
return 0;
}

```

Output:

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● Enter number of disk requests: 8
Enter disk requests:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Enter disk size (maximum cylinder): 200
Enter direction (L for Left, R for Right): R

Seek Sequence:
53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 37 -> 14

Total Seek Distance = 331
Average Seek Distance = 41.375

```